

```
dovecot:x:97:97:dovecot:/usr/libexec/dovecot:/sbin/nologin
vbg1:x:501:501:vbg1:/home/userhome/vbg1:/bin/bash
websms:x:502:502:websms:/home/userhome/websms:/bin/bash
rajveer:x:503:503:rajveer:/home/userhome/rajveer:/bin/bash
testrpm:x:504:504:testrpm:/home/userhome/testrpm:/bin/bash
[root@rdso ~]#
```

This shows that the required changes have been made. Once this is done, the following remains to be completed:

- On the file server, individual folders for all users need to be created and exported;
- On the clients, suitable mountpoints need to be created and NFS exports need to be mounted.

Let us create the home directory for these users in the shared export folder on the NFS server:

```
[root@vbg ~]# mkdir /myusersdata/vbg1
[root@vbg ~]# mkdir /myusersdata/rajveer
[root@vbg ~]# mkdir /myusersdata/websms
[root@vbg ~]# mkdir /myusersdata/testrpm

[root@vbg ~]# ls -l /myusersdata/
total 16
drwxr-xr-x 2 root root 4096 Oct 21 00:14 rajveer
drwxr-xr-x 2 root root 4096 Oct 21 00:15 testrpm
drwxr-xr-x 2 root root 4096 Oct 21 00:08 vbg1
drwxr-xr-x 2 root root 4096 Oct 21 00:14 websms
[root@vbg ~]#
```

Now that you have the users, you just need to mount the corresponding folders in the client. To get the home directory for the user vbg1 at the right location, as per the output of the `getent passwd` command, i.e., `/home/userhome/vbg1`, mount the NFS exported folder `/myusersdata` to the mountpoint `/home/userhome` in the client machine:

```
[root@rdso ~]# mkdir /home/userhome
[root@rdso ~]# mount 192.168.122.1:/myusersdata /home/userhome
[root@rdso ~]# df -h

Filesystem                Size      Used Avail Use% Mounted on
/dev/mapper/VolGroup00-LogVol00
                           8.6G      5.3G   3.0G   64% /
/dev/hda1                  99M       13M    82M   14% /boot
tmpfs                     250M           0  250M    0% /dev/shm
192.168.122.1:/myusersdata
                           19G       17G   1.6G   92% /home/userhome

[root@rdso ~]#
```

On the client, log in as user vbg1 and see the results =>

```
[root@rdso ~]# su - vbg1
-bash-3.2$ pwd
/home/userhome/vbg1
```

Your job is primarily done. But there are various problems with this approach:

- The entire folder `/userhome` is mounted on this system and the data of all the users is visible to all, as can be seen from the following command:

```
[root@rdso ~]# ls /home/userhome/
rajveer testrpm vbg1 websms
```

- To avoid the above, if we were to just mount a specific user's folder, then we would have had to create individual folders for each user. That would be a nightmare for the management. Every time a new user joined the organisation, you would need to create a new mountpoint (`/home/userhome/xyz`) in all the client machines.

Auto-mount is an elegant way of dealing with this problem. As the name (auto-mount) suggests, it will automatically mount filesystems for you when they are required, and unmount them when not required. Automounter is implemented through autofs in RHEL systems. The RPM installed in my system is `autofs-5.0.1-0.rc2.131.el5`. Generally, auto-mounter is configured through text files -- `auto.master` and `map file(s)`.

The default configuration file, `auto.master`, defines mount point(s) under which filesystems are automatically mounted on the client machine and the `map file(s)` that are associated with these. As an example, the first map defined in the default `/etc/auto.master` on my client looks like what's shown below:

```
/misc /etc/auto.misc
```

This basically means that for all automatically mounted partitions under `/misc`, the definitions are provided by the `/etc/auto.misc`. Let's create our own entry:

We have to auto-mount folders as `/home/userhome/[xyz]`, where `[xyz]` represents the name of the logged in user. Therefore, our top folder in the mount hierarchy is `/home`. Hence, you can add the following line to `/etc/auto.master` and comment the other lines =>

```
/home /etc/auto.home
```

Now, let us create the map file `/etc/auto.home`. This map contains the parameters found in the mount command. If we, use the following command to manually mount the NFS partition:

```
[root@rdso ~]# mount 192.168.122.1:/myusersdata /home/userhome
```

then our `/etc/auto.home` shall look like =>

```
[root@rdso ~]# cat /etc/auto.home
userhome -rw 192.168.122.1:/myusersdata
```